

Customer Portal App Design

Exercise

Outline

In this exercise, we will use the Architecture Design Process to design the Architecture for a Customer Portal application. We will follow the three steps of the design process:



At the end of this exercise, we will have the Architecture Blueprint for this use case. The solutions will be presented and discussed in the Boot Camp.

Hands-on

Design the Architecture for the Customer Portal Use Case, using the 3 phases of the Architecture Design Process. This exercise will be done in groups of 3–4 people.

Use Case – Customer Portal

The ACME company wants to build a Customer Portal to consolidate all SAP and Salesforce customer information into a single application to support the agents on the field. In this portal, agents can view the customers who fall under the purview of their territory and schedule meetings with them. The administrators can manage agents, locations, customers, and their mappings to each other. The users are authenticated via the company's Azure AD.

Functional Requirements

Customer Information

- Information about customers is shown in the Customer Portal, including map location.
- All information can be edited except for the customer number.
- Once the customers are onboarded, they need to digitally sign a contract using DocuSign which is stored in S3

Integration with SAP

- Integration with SAP is required for retrieving and updating customers' and orders' information.
- It uses 2 BAPIs to retrieve the information from Customers and Orders, and a web service to update changes done on the Customer Portal.

Integration with Salesforce.com

- Integration with Salesforce.com is required for retrieving customer opportunities.

Aggregate the Information

- Data from SAP and Salesforce.com must be aggregated in the application
- Prospects and potential customers aren't created in SAP, so the Customer Portal will only show information for actual customers.

Help

- The application should provide online help to the users.
- The help contents should be editable within the application.

Audit

- All information changes made by users or synchronization processes must be logged for auditing purposes

Notifications

- The customer will be able to be notified via SMS, Email, or WhatsApp. The company plans to use its Microsoft 365 APIs for email sending
- The application will be able to send a Google calendar invite to the prospective customer for a possible meetup with the agents

Exercise Requirements

- Use the Architecture Design Process presented in class
- Prepare a draft of your concepts first while you iteratively Disclose, Organize, and Assemble your solution
 - Make sure you provide a list of identified concepts first before thinking about applications
 - Then, for the Organize step, start placing those concepts in the slide deck template organized into bounded contexts
 - In the Assemble step, place each bounded context in the correct application type and make sure you have the team(s) organization and independence in mind
 - Name each application appropriately following the naming best practices
- At the end, present the final architecture blueprint and discuss it with the trainer and other participants